

Ginevra Cerri

SENIOR PERCEPTION ENGINEER
ROBOTICS · COMPUTER VISION · AUTONOMY

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I give robots and vehicles their sense of sight, and carry it from research prototype to shipped product. Today I run the perception workstream inside a larger autonomy stack: object detection and tracking, depth and spatial estimation recovered from cameras alone, and vision-language models that identify what the scene contains, integrated with the modules around it.

01 EXPERIENCE

Five years turning perception research into systems that run on real vehicles.

2nd

place at A2RL, the team's first eligible autonomous racing competition

250 km/h

the speed the perception stack has to stay reliable at

5+

engineers led on an external delivery, end to end

04 CORE SKILLS

PERCEPTION & ML

- Computer vision
- Object detection & tracking
- Monocular depth estimation
- Sensor fusion
- Vision-language models
- Model optimization

ENGINEERING

- Python
- C++
- ROS
- Docker
- SQL
- MATLAB & Simulink
- Git

LANGUAGES

- Italian — native
- English — C2
- French — B2
- Spanish — B1

02 EDUCATION

Milan, Rome and a semester in Paris.

Senior Perception Engineer · SIRB AI

Jan 2026 — Present · Abu Dhabi

Leading perception for swarm autonomy, moving from research to a commercially deployed product.

- Architecting the end-to-end perception framework for the transition from research to product: computer vision model optimization, deployment targets, and a large-scale data architecture for continuous retraining.
- Owning 4 stack sub-components for vision and navigation tasks, one of them with an external startup, while coordinating with the other module leads for integration.
- Applying vision-language models to classify the identity and affiliation of detected entities in the scene, alongside spatial estimation that raises object localization accuracy without depth sensors.
- Designed field demos with the other sub-team leads, then validated and presented them on site, representing perception directly to clients.

Senior Perception Engineer · Technology Innovation Institute

Oct 2023 — Dec 2025 · Abu Dhabi

Leading perception for TII Car Racing team, racing a driverless car wheel-to-wheel above 250 km/h.

- Led the Perception Division to a **2nd place finish in the team's first eligible A2RL competition**. Opponent detection and tracking, visual localization error estimation inside a hard latency budget.
- Managed a 5+ person external team on a custom perception algorithm: defined requirements, built the dataset, benchmarked performance and drove it to delivery.
- In the competition's earliest stage, built and benchmarked the car's sensor suite, then delivered it with a baseline perception stack to the organizers, acting as technical consultant.

Customer Solutions Engineer · Unchained Labs

Dec 2024 — Sep 2025 · Remote

- Directed end-to-end development of the Hunter ecosystem, an AI agent for on-chain insights, covering product scoping, roadmap, design direction and release, alongside a full-time engineering role.

Research Assistant, Robotics & CV · Polytechnic of Milan

Sep 2022 — Jul 2023 · Milan

- Kicked off, designed and deployed the perception stack for the Aida project, which spun off into Niulinx, an autonomous-driving startup for urban mobility.
- Collaborated with Maserati S.p.A. on 1000 Miglia 2023, running the first autonomous vehicle in the race's history.

Undergraduate Researcher, CV · IZONICS

Jan 2022 — Jun 2022 · Paris

- Labelled 100+ samples into a training dataset for a CNN reaching 95% accuracy classifying biological vs. PVC particles; reported results directly to the founders.

MSc Automation & Control Engineering

2020 — 2023 · Milan, Italy

Polytechnic of Milan

Graduated 110 cum laude/110. Thesis nominated for the Award for Best Thesis on Advanced A&C Techniques.

Erasmus+ Exchange Programme

2022 · Paris, France

ESIEE Paris, Grande École d'Ingénieurs

BSc Mechanical Engineering

2016 — 2020 · Rome, Italy

La Sapienza, University of Rome

03 SELECTED PROJECTS

Two shipped **on the road**, two shipped **on the web**.

A2RL Autonomous Racing

2024–25

Detection, tracking and opponent state estimation. 2nd place, debut season

Hunter

2024–

AI agent turning raw on-chain activity into readable insight. I own product and release.

1000 Miglia 2023

2023

On a Maserati MC20 Cielo, the first autonomous vehicle to run the historic road race

Kymatics

2024

Voice-to-deployed-web-app code assistant, model agnostic. Built at Mistral Worldwide Hacks.